

Assignment 1

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1.1 Strategic asset allocation

This report analyzes the historical performance of a globally diversified portfolio from the perspective of an EUR-based investor. The dataset includes 20 financial instruments across developed and emerging markets, encompassing equities, sovereign bonds, and regional exposures. In line with standard market practice, foreign currency risk was hedged for bonds but not for equities.

All returns were converted into EUR using the appropriate monthly spot exchange rates. For bonds denominated in non-EUR currencies, returns were adjusted for carry by adding the monthly return of the relevant FX carry series. Monthly returns were then computed as simple percentage changes in the EUR-adjusted total return indices.

At the asset level, we observe significant variation in both return and risk characteristics (Table 1). Among equities, the S&P 500 (S&PCOMP) stands out with the highest annualized return (8.05%) and a Sharpe ratio close to 0.50, confirming its dominant long-term risk-adjusted performance. Conversely, indices such as TOPIX and the IG bond Sterling Corporate show low or modest returns coupled with high volatility, resulting in weak Sharpe and Sortino ratios. On the bond side, several EUR-denominated instruments (e.g., Italy and Spain government) display exceptionally high Sharpe ratios (above 1), low drawdowns, and strong downside protection. This performance is mainly driven by their low volatility and defensive nature as core fixed income exposures, rather than by any currency hedging effects.

Based on these individual assets, we constructed an equally weighted portfolio, with each of the 20 instruments receiving a constant monthly weight. The resulting portfolio achieves a 5.90% annualized return with a volatility of 9.94% translating into a Sharpe ratio of 0.59 and a Sortino ratio of 0.82. These risk-adjusted metrics (Table 2) indicate a balanced profile, capturing upside return while mitigating downside risk. Compared to the individual assets, the portfolio offers smoother performance and lower tail risk. Notably, the maximum drawdown is contained at -29.3% lower than many of the single instruments, particularly in equity markets where drawdowns often exceed 50–60%. This reinforces the benefit of combining uncorrelated exposures across asset classes and regions.

The cumulative portfolio performance (Figure 1) shows sustained growth, with an initial EUR 1 investment reaching a value of over EUR 7 by 2024. While major crises such as the 2008 financial collapse and 2020 COVID shock caused temporary losses (Figure 2), the portfolio recovered strongly in both instances. The quick rebounds highlight the structural diversification built into the allocation. In conclusion, these findings illustrate that even a simple equal-weighted allocation, when properly converted and hedged, can yield competitive and resilient performance. The combination of equities and bonds, across developed and emerging markets, provides a robust foundation for long-term EUR investors, both in terms of return potential and downside protection.

Metric	DJEURST	IBEX35	S&PCOMP	FTSE100	TOKYOSE	MSFXJIPS	MSEMFAS	MSEFLAS	MSEMAFS	MLBDAML	MLTAML	MLESAML	MLTRSM	MLUKAML	MLEXPTL	MLCORM	MLURORL	MLHEUCL	MLHMACL	MLEMGBL
Annualized Return (%)	4.69	3.60	8.05	3.65	0.23	3.91	4.08	8.30	2.31	1.32	6.68	5.67	4.81	3.35	3.77	5.93	3.19	4.68	7.21	8.68
Annualized Volatility (%)	17.53	20.67	16.25	15.23	20.75	19.77	23.32	29.95	22.04	4.15	5.78	5.71	10.18	9.42	3.98	9.98	10.12	11.54	11.77	14.32
Sharpe Ratio	0.27	0.17	0.50	0.20	0.01	0.20	0.17	0.28	0.11	1.04	1.16	0.99	0.48	0.36	0.95	0.59	0.31	0.41	0.61	0.61
Peak	5.00	5.64	15.66	3.45	1.19	3.98	5.05	25.33	3.33	5.39	11.12	6.80	5.95	4.59	3.21	8.04	3.00	3.48	11.41	15.26
Max Drawdown	-0.60	-0.62	-0.66	-0.62	-0.69	-0.58	-0.75	-0.58	-0.59	-0.21	-0.20	-0.20	-0.25	-0.33	-0.17	-0.25	-0.41	-0.40	-0.31	-0.39
Downside Deviation (%)	12.67	14.11	11.03	10.51	12.79	13.97	15.81	21.51	17.05	2.76	4.19	4.00	4.71	6.23	3.39	4.73	7.62	10.80	6.84	11.88
Sortino Ratio	0.37	0.25	0.73	0.29	0.02	0.28	0.26	0.39	0.14	1.57	1.59	1.42	1.03	0.54	1.11	1.26	0.42	0.43	1.06	0.73
Calmar Ratio	0.08	0.06	0.12	0.05	0.00	0.07	0.05	0.14	0.04	0.20	0.34	0.28	0.20	0.10	0.23	0.24	0.08	0.12	0.23	0.22

Table 1: Performance metrics across global asset classes

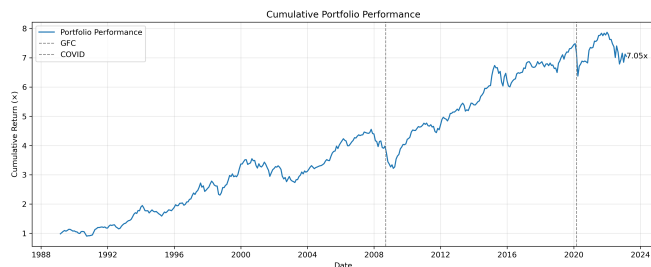


Figure 1: Cumulative portfolio performance

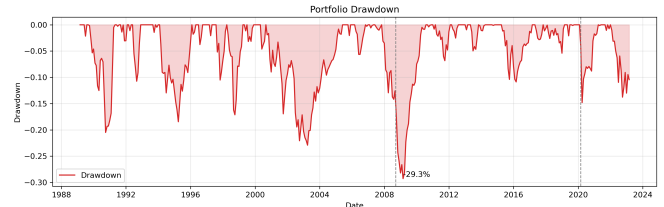


Figure 2: Portfolio drawdowns

	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	Downside Deviation (%)	Sortino Ratio
Equally Weighted Portfolio	5.90	9.94	0.59	7.16	0.82

Table 2: Summary statistics for the equally weighted portfolio

Given the converted and hedged returns for each of the assets, the optimal strategic asset allocation (SAA) over time to overall equities, bonds, and credit was found based on an annual target risk of 10%. The optimization was computed using data from 1998 onwards, as data for some assets were missing in the earlier years. For each month, the expected

return was calculated as the rolling average over the previous 12 months. For the risk model, zero correlation between the assets was assumed and volatility was estimated with an exponentially weighted moving average (EWMA), using a lookback window of 12 months. The portfolio’s expected return was then maximized under a 10% annualized volatility target, with no shorting or leverage at either the asset or portfolio level. We run the optimization at year-ends and forward filled the weights afterwards. Table 3 summarizes selected non-zero portfolio weights.

When comparing to the simple equal-weighted portfolio, the optimized portfolio performed worse (Table 4). A possible reason for this could be that the constraints imposed on the optimization (e.g., no shorting, no leverage, annualized volatility target) might have limited the ability to achieve higher returns. These constraints reduce risk exposure but could also hinder the portfolio from taking advantage of favorable market conditions. Additionally, the assumption of zero correlation and the use of EWMA for volatility estimation might not accurately capture the true risk dynamics of the assets. In reality, assets are often correlated, and the risk model might have oversimplified the portfolio’s exposure. The choice of a 12-month rolling window for estimating expected returns and volatility might also have been suboptimal for capturing the true risk and performance of the assets, particularly during periods of market change or volatility. Finally, the market conditions during the period of analysis might have favored a more diversified approach, such as an equal-weighted portfolio. The equal-weighted portfolio maintained broad exposure, while the optimized portfolio became overly concentrated in a few assets during certain years, leading to higher volatility and potential underperformance.

Furthermore, the optimized portfolio relies heavily on in-sample estimates of expected returns and volatility, which may not generalize well out-of-sample. This introduces model risk and makes the portfolio highly sensitive to estimation error, especially in periods of regime shifts or structural market changes.

Year	Instruments with Non-Zero Weights
2000	TOKYOSE (29.3%), MSPXJP\$ (9.9%), MSEMFA\$ (12.1%), MSEFLA\$ (7.3%), MSEMAF\$ (13.8%)
2005	MSEFLA\$ (47.4%), MSEMAF\$ (52.6%)
2010	MSPXJP\$ (12.7%), MSEMFA\$ (12.5%), MSEFLA\$ (26.6%), MSEMAF\$ (8.4%), MLHEUCL (30.4%), MLHMACL (9.3%)
2015	S&PCOMP (100.0%)
2020	DJEURST (16.3%), S&PCOMP (81.4%), TOKYOSE (2.2%)

Table 3: Non-zero portfolio weights for selected 5-year intervals

	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	Downside Deviation (%)	Sortino Ratio
Optimal SAA	3.27	14.45	0.23	10.37	0.31

Table 4: Summary statistics for the optimal SAA

1.2 The case of NBIM fund

The Norges Bank Investment Management (NBIM) sovereign wealth fund was established to manage the financial wealth generated by the country’s oil revenues. The fund’s strategic asset allocation (SAA) is set by a benchmark portfolio made up of 70% equities and 30% fixed income. The goal of this allocation is to maximize real returns over the long term while keeping the level of risk moderate. Tactical Asset Allocation (TAA) allows NBIM to make short-term changes to its investments based on current market conditions. The goal is to improve returns by taking advantage of market opportunities as they arise. By doing this, the fund keeps a tracking error limit of 1.25% to make sure that these short-term adjustments don’t stray too far from the benchmark’s overall risk.

NBIM aims to maximize long-term real returns after costs while effectively managing risk within acceptable levels. To achieve these objectives, NBIM operates within several key constraints that guide its SAA and TAA. The fund is subject to Legal Limitations, which prohibit the use of leverage and short-selling. The Responsible Investment Guidelines followed by NBIM exclude companies that violate ethical standards, ensuring the fund aligns with sustainability goals. NBIM also adheres to Liquidity Requirements, prioritizing investments in liquid assets to maintain flexibility and the ability to respond to market changes, limiting investments in illiquid assets such as real estate (max 7%) and renewable energy (max 2%).

The Norwegian state’s goals for NBIM fund are closely aligned with its asset allocation strategy. NBIM’s portfolio is well diversified, which helps generate higher returns and stability. Adherence to Responsible Investment Guidelines, excluding companies involved in unethical activities, also allows fund to align with state’s sustainability goals.

The fund’s exposure to non-liquid assets is kept small to maintain liquidity for market changes and policy-driven withdrawals. Non-liquid assets are more volatile and harder to value, so limiting their exposure helps manage risk and maintain portfolio stability. Illiquid assets also come with higher transaction costs and management fees, making them less cost-effective. Keeping these assets limited helps the fund operate more efficiently.

Interestingly, although NBIM is funded by oil revenues, it actively pursues environmental objectives. It excludes companies that violate environmental standards, invests in green infrastructure, and engages in active stewardship. This apparent paradox is actually a conscious strategy: using wealth generated from fossil fuels to support a sustainable transition. The fund represents a pragmatic balance between financial prudence, long-term growth, and ethical responsibility.